

Niklas Virtanen

FC Midtjylland sport scientist · Catapult consultant · 60 min consultation

GOAL: Extract 3 ideas that make MicroPulse meaningfully better. **Not selling.**

If he gives 3 real insights, he just saved you 3–6 months of experimentation. That's the entire value.

Before the meeting (1 hour prior)

- **Laptop** running localhost, logged in as Breiðablik coach. Demo players: Höskuldur, Kristófer, Óli Valur.
- **Printed list of questions** (this sheet) — don't trust your memory.
- **Notebook + pen** — handwritten notes work better than typing.
- **Coffee/water** offered.
- **No slides** — show the live system only.

0:00 · Set the tone (3 min)

Say to him directly:

"You know IMA and Catapult deeper than I do. I want 45 minutes of unvarnished feedback — where am I thinking about this wrong, where are the holes in the design, where should I improve. Not a sales pitch, just consultation. I'm going to ask hard questions — don't soften your answers."

This makes him relaxed and opens the door to blunt feedback.

0:03 · Section 1 · Threshold tuning (10 min)

► **Show:** Decision Summary card with Sprint Exposure chip + Sprint Speed Drop chip + Bishop CoD chip.

Q1. *We use 18 km/h for IMA Free Running band 5 (Catapult default). Bradley 2018 recommends 19.8 km/h, Buchheit 2014 uses 18.0. What does FCM use and why?*

Q2. *Sprint Exposure (Malone 2018) Safe range — the original paper was 80–130%. We widened it to 80–180% after empirical tuning on Icelandic football weeks (a typical healthy week delivers 130–160% of one match). Defensible or am I missing something?*

Q3. *Sprint Speed Drop > 10% (Edouard 2019). Should this be per-player? Are we losing signal by using a population threshold when individual variance is high?*

→ **Listen for:** Specific numbers FCM uses. Whether threshold differences are position-tied or data-driven.

0:13 · Section 2 · CoD asymmetry framework (10 min)

► **Show:** IMA Intelligence page → expand Höskuldur's row (64% high-tier asymmetry on May 12).

Q1. *Bishop 2020 sets 15% as the threshold. Does FCM use 15% or something different? Have you seen false positives at 15%?*

Q2. *We show the 3 tiers (low / medium / high) separately in the drilldown. Low + medium asymmetry we don't colour-flag (positional habit). Right call, or should medium tier also flag?*

Q3. *When should a coach act on 64% high-tier asymmetry like Höskuldur here? Immediately with a Copenhagen adductor block, or first monitor for 3–5 days to rule out daily fluctuation?*

→ **Listen for:** His empirical experience on asymmetry-magnitude → action correlation.

0:23 · Section 3 · Sprint Exposure GPS fallback (10 min)

This is the newest work — he hasn't seen it.

► **Show:** Höskuldur's Sprint Exposure chip showing 'Based on 1 of 5 matches (others missing data)' + walk him through the calibration algorithm.

Q1. *Catapult Free Running config was disabled on older activities — IMA bands are 0 but GPS V5+V6 efforts are present. We estimate IMA strides from GPS efforts × per-player ratio (calibrated from one measured match). Is this legitimate, or am I adding noise?*

Q2. *Per-player ratios: Höskuldur 10.8 strides/effort, Kristófer 4.6, Óli Valur 5.8 — 2× spread. Does FCM keep individual ratios like this in your models, or is that too high-variance for generalisation?*

Q3. *I only use match-day calibration (training cadence ratios are too variable). Agree, or overly conservative?*

→ **Listen for:** Whether the methodology is methodologically sound or overengineered.

0:33 · Section 4 · Per-position individualization (8 min)

Our biggest gap — acknowledge it upfront.

► **Show:** Player list in Decision Summary with mixed positions visible.

Q1. *Our biggest weakness today is population-norm thresholds — same flag for a CB and a winger. Does FCM run per-position tuning?*

Q2. *How do you calibrate per-position thresholds? 30-m sprint test + ForceDecks profile + position grouping, or something more sophisticated?*

Q3. *If I wanted to start per-position tuning in MicroPulse — which 1 metric would make the biggest difference to individualise, and which 1 would be a waste of time?*

→ **Listen for:** His actual workflow at FCM. Not theory — practice.

0:41 · Section 5 · Decision flow + override-aware engine (8 min)

► **Show:** Daily Briefing dashboard → click into Decision Summary modal → verdict explanation.

Q1. *Our verdict engine is deterministic — AI (Claude Haiku) is only a refinement layer on the ELITE tier. The verdict decision runs in code with cite-able reasoning. Does FCM trust AI or deterministic systems more for injury-risk decisions?*

Q2. *Coach override flow — if the coach says 'FULL' but the engine says 'REDUCED', we surface an override-conflict warning. How does FCM handle coach-vs-engine disagreement?*

Q3. *When a player gets injured DESPITE a 'safe' verdict, how does FCM avoid hindsight bias in the post-hoc review? Do you have a process for verdict-vs-outcome calibration?*

→ **Listen for:** His process for model improvement loops.

0:49 · Section 6 · Open-ended wisdom (10 min)

No specific UI — just questions. These are the gold-mine questions.

□ **THE GOLD QUESTION:** *"What 1 or 2 metrics has FCM dropped from your model because they didn't work in practice? Something that looked good in the paper but flopped in daily use."*

Q2. *What metric does FCM wish Catapult exposed but they don't?*

Q3. *If you were starting MicroPulse from scratch with 6 months — where would you focus? What's overrated in analytical tooling right now?*

Q4. *What mistakes have you seen teams make in the implementation phase that you'd warn me about?*

Q5. *If I could call you once every 3 months with one question — what's the most valuable question I could ask?*

→ **Listen for:** These answers shape 80% of the future roadmap.

0:59 · Closing (3 min)

Before you stand up, tell him:

| *"Before I get up — let me play back the 3 things I heard: ..."*

Repeat in your own words:

- 1 metric he dropped
- 1 metric he wishes Catapult exposed
- 1 idea he disagrees with you on

Two effects: (1) confirms you heard him right, (2) shows him you were actually listening. He'll be more likely to keep giving you advice later.

Thank him. Ask permission to follow up with one email within the week.

Within 30 minutes after the meeting

Sit down. **Don't open the laptop.** Write on paper:

1. **Top 3 insights** he gave that you didn't know before
2. **1 dropped metric** + why (the most valuable item)
3. **1 missing Catapult metric** per his view
4. **2–3 verbatim quotes** — something you want to remember exactly
5. **Open follow-up questions** — things you forgot to ask or he didn't fully answer

Within 24 hours: convert these into MicroPulse roadmap items in codebase comment headers or Notion. Otherwise 70% of the insight evaporates within a week.

Within 1 week — follow-up email

Keep it short (5–7 lines):

- Thank him again
- Two specific ideas you're starting on (shows him he influenced you)
- 1 follow-up question you wanted to add
- Offer to show him the system once you've implemented something he suggested

This keeps him invested. Sport scientists like Niklas are often glad to see their ideas implemented in real systems — much more rewarding than publishing a paper nobody reads.

What NOT to do

❑ Avoid	❑ Instead
Explain Malone 2018 to him (he knows it)	Just cite the threshold and ask his view
Sell Lite / Pro / ELITE tier pricing	This isn't a sales meeting
Show Breiðablik pilot statistics	He doesn't care — show the architecture
Use the word "AI" repeatedly	Sport scientists are tired of that word
Defend chosen thresholds	Just say "we set it here, what do you think?"
Have an answer ready for every question	It's OK to say "I don't know — good point"

Note-taking template (use this in your notebook)

Threshold FCM uses	_____
Per-position approach	_____
Dropped metric	_____
Missing Catapult metric	_____
Coach override handling	_____
Verbatim quote 1	_____
Verbatim quote 2	_____
Top idea to implement	_____

Open follow-up Q	
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